Violence Detection in Surveillance Videos Using a Hybrid CNN–LSTM Architecture

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***Abstract*—The surveillance of violent activities using CCTV footage is a complicated issue since a huge amount of video information is produced and manual surveillance has limitations. To solve this problem, this project proposes the automated system of violence detection that will analyze the surveillance video footage through the deep learning methods. The suggested suitable method will utilize both a convolutional neural network (CNN) to extract the spatial features of an image and a long short-term memory (LSTM) network to compute the temporal trends of succession frames of a video. MobileNetV2 is selected as the backbone CNN because it has a lightweight structure and is also computationally efficient, hence suitable in surveillance. The system is tested and trained using the RWF-2000 data that composes actual CCTV camera footage of violent and non-violent cases. The model is able to learn to discriminate violent activities using both appearance and motion information by working out the video sequences of fixed length frame windows. The results of the experiment prove that the offered CNN-LSTM architecture can be successfully applied to the detection of an aggressive event in the surveillance video, which also shows its prospect of being a useful instrument to support security surveillance.**

***Index Terms*—Violence Detection, Deep Learning, CNN-LSTM, Surveillance Systems, Action Recognition, MobileNetV2, Real-Time Processing, Computer Vision**

I. Introduction

Closed-circuit television (CCTV) is a popular surveillance method of controlling areas both in public and private to increase safety and security. Nonetheless, it is not practical to subject surveillance footage to constant human observation since there are numerous cameras and the surveillance system needs human operators to invest a lot of time. Because of this, critical occurrences like violent occurrences can be missed or realized when they are already in existence. Surveillance data analysis through automated means has thus become a research topic in intelligent security systems [10], [15] since it is now an active field of research..

Automated violence recognition in surveillance videos is an attempt to help the monitoring systems by examining video streams and detecting activities that might be regarded as aggressive. Compared to the conventional image-based models, violence detection does not only presuppose the visual look but also the movement pattern through time. Violent activities are usually sequential and require time analysis as a fundamental part of successful detection. Moreover, real-world surveillance camera images pose other problems like change in light systems, angles, and occlusions, and cluttered backgrounds that increase the difficulty in reliable classification [1]. More so, surveillance camera images on the ground also have problems like changing light systems, angle, and occlusions, as well as cluttered backgrounds that make it hard to achieve reliable classification..

A. Problem Statement

The main problem of automated violence detection is to be able to detect the violent behavior and correctly identify it with the use of video sequences of uncontrolled surveillance. This dilemma can be summarized in the following way:

Temporal Understanding: Violent activities are dynamic and change with a series of frames. One frame may not be enough to tell the difference between violence and other non-violent activities that look similar to it. Hence, to be robustly detected, modeling of temporal relations between sequential frames with the help of spatiotemporal learning approaches is necessary [5], [6].

Computational Efficiency: Surveillance systems normally run on small computational resources. The detection methodology should be effective to handle video data without having excessive computation load that could not give good performance. MobileNet (1.5) and MobileNetV2 are lightweight deep learning architectures that have been offered to overcome this problem [8], [9].

Environmental Variability: CCTV shots differ greatly in terms of the layout of the scene, lighting, position of the

camera and the density of the crowd. These variations must be generalized well by a robust violence detection system rather than having to be sensitive to a particular environment [15].

B. Primary Objectives

The main aim of the project is to model and install an automated violence detection system that will be able to process CCTV video records and detect violent actions through the techniques of deep learning. The work is concentrated on the acquisition of spatial and time-related features of violent behavior on the basis of video information.

The objectives of this project are as follows:

- To identify spatial information, aiming to find lightweight convolutional neural network elements, in each frame of the video [9].
- To predict the interdependences of time in a stream of successive frames of a video sequence with a recurrent neural network [6].
- To categorize video sequences into violent and non-violent based on learned spatiotemporal features.
- To assess the strategy of the proposed approach with the help of a real-word surveillance video dataset [7].

C. Principal Contributions

The primary findings of the work are as follows:

- Design of a hybrid CNN-LSTM framework that integrates spatial feature extraction and temporal sequence modeling of violence in CCTV video due to recent CNN-RNN based designs [11].
- MobileNetV2 is an efficient feature extractor that should be used in surveillance situations with limited computing capabilities, as it is suitable in this case, as well [9].
- Adoption of video sequence-based classification model which can utilize temporal context to enhance recognition of violence over single frame analysis.
- The given model is to be tested experimentally on the RWF-2000 dataset to show that it can effectively work with real-life surveillance videos [7].

D. Paper Organization

The rest of this paper will be structured in the following way. Section II is a review of related work in action recognition and violence detection. Part III outlines the methodology offered, which consists of preprocessing datasets and model structure. Experimental results and analysis are in section IV. Section V talks about limitations and future work, and Section VI provides a conclusion of the paper.

II. Related Work

The process of automated violence detection is directly associated with the larger area of video-based action recognition. Initial work has been done using handcrafted features and traditional machine learning as well as recent work has mostly used deep learning models that can learn spatiotemporal representations directly off of video data without requiring handcrafted features [1], [10], [15].

A. Action Recognition Foundations

In the past, manual methods of action recognition worked with manually constructed spatiotemporal features including Space-Time Interest points (STIP) [2], histogram of oriented gradient (HOG) and optical flow descriptors [3]. Although these techniques performed reasonably well under controlled conditions, they were very sensitive to changes in lighting, viewpoint and complexity of backgrounds which constrained their usefulness in real world surveillance.

More precise motion information is captured by two-stream convolutional neural networks that take RGB frames and optical flow input images separately and process them separately [4]. Even though useful to recognize actions, the need to explicitly compute optical flow made these methods more expensive, in terms of computation, and thus not very applicable to real-world surveillance.

B. Deep Learning for Violence Detection

Spatial feature extraction and temporal modelling combine to offer better visibility of violence when using deep learning procedures. Three-dimensional convolutional neural networks (C3D) naturally generalize the 2D convolutions into the temporal axis, making it possible to directly learn spatiotemporal cues in video sequences [5] through learning 3D filters instead of 2D onesThree-dimensional convolutional neural networks (C3D) are a generalisation of the 2D convolutions into the time dimension, where instead of learning a 2D filter, spatiotemporal features are learned directly by learning a 3D filter. Nevertheless, they are too complex to be used in resource-limited settings

Another and more popular method is that which uses convolutional neural networks together with recurrent neural networks. Specifically, LSTM networks can be used to represent the temporal violations between successive video frames and work well in modeling their interaction with the network of video frames, in general, and memorization in particular (Long and Short-term memory) [6]. CNN-LSTM models have also been used to show good results in violence detection where appearance and motion are jointly captured as a unit of information to predict violence occurrences successfully [11].

Convolutional LSTM (ConvLSTM) networks further integrate convolution operations within recurrent structures to preserve spatial information while modeling temporal dynamics [12].

More recently, transformer-based architectures such as Video Vision Transformers (ViViT) have been proposed to model long-range temporal dependencies more effectively [14]. While these approaches achieve strong performance, they generally require large-scale datasets and significant computational resources.

C. Surveillance-Specific Datasets

Authentic data sets are the key to advancements in the study of violence detection. In the RWF-2000 data set, there were 2000 actual CCTV videos which depict violent and non-violent scenes. These videos address such difficulties as

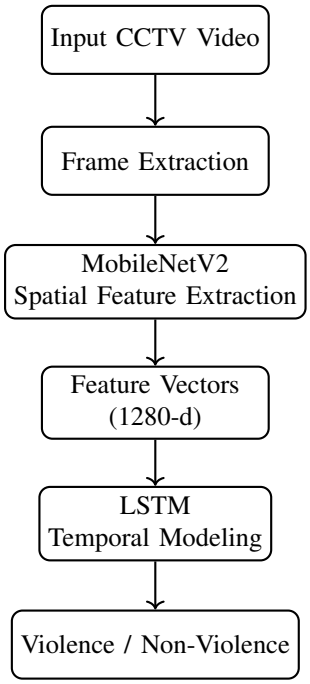

Fig. 1. CNN-LSTM violence detection system architecture.

camera angles, the light variations, and barriers [7]. RWF-2000 is normally used to test violence detection systems under surveillance, due to its resemblance to a real life scenario.

There are also other datasets that have been developed in aid of violence detection studies, however most of them are specific to a particular area and lack diversity. [13].

D. Lightweight Deep Learning Architectures

The fact that the system is fast is important in surveillance. The depthwise separable convolutions in MobileNet [8] reduce the size of the model and make it effective at the same time. MobileNetV2 is even more desirable since it applies inverted residual blocks and linear bottlenecks thus it can be utilized in video operations where a lightweight and fast feature extractor is required [9].

Light CNN backbones are also required particularly to implement violence detection systems in real time surveillance.

III. Methodology

This part discusses our method of violence identification in CCTV videos. It discusses the process of cleaning the data, the spatial features, the time aspect modeling, and decision at video-level.

A. System Architecture

The general plan of the violence detection system is depicted in Figure 2. The pipeline begins by taking input video sequences then derives and preprocesses frames. A frame-based image -recognition network extracts spatial features. A recurrent model is a model that represents how those features vary with time. The last model will provide the yes/no response on whether the video is violent.

B. Dataset Preprocessing

1) RWF-2000 Dataset: The RWF-2000 dataset that consists of 2,000 CCTV clips depicting violent and non-violent scenarios is used to train and test the system. The clips will be divided equally between both classes and will be separated into training and testing sets.

2) Frame Extraction: The surveillance videos in the dataset are of varying frames. In order to maintain a time-based aspect, we divided the videos at a constant rate:

$$f_{extract}(V) = \{I_i \mid i = 0, \Delta f, 2\Delta f, \ldots\} \tag{1}$$

where V represents the input video, I_i denotes the extracted frame at index i, and Δf is the sampling interval. In this work we use a frame rate of 10 frames per second.

$$fps_{target} = 10 \Rightarrow \Delta f = \left\lfloor \frac{fps_{original}}{fps_{target}} \right\rfloor \tag{2}$$

This technique reduces redundancy of information and retains sufficient movement information to identify violence.

3) Temporal Sequence Formation: Violent acts span various frames, and they require an idea of time to be properly identified. Our frames are extracted into constant length sequences:

$$S_j = \{I_{j \cdot s}, I_{j \cdot s+1}, \ldots, I_{j \cdot s+T-1}\} \tag{3}$$

where $T = 16$ denotes the sequence length and s is the stride. A 50% overlap ($s = 8$) is used to capture motion continuity across sequences and improve learning robustness.

4) Frame Normalization: All the frames are scaled down to fit the size of the MobileNetV2 anticipated input:

$$I_{norm} = \text{Resize}(I, 224 \times 224) \tag{4}$$

$$I_{tensor} = \frac{I_{norm}}{255.0} \tag{5}$$

$$I_{final} = \frac{I_{tensor} - \mu_{ImageNet}}{\sigma_{ImageNet}} \tag{6}$$

where $\mu_{ImageNet}$ and $\sigma_{ImageNet}$ denote the mean and standard deviation values computed from the ImageNet dataset. This normalization ensures stable training and effective transfer learning.

C. Spatial Feature Extraction

1) MobileNetV2 Architecture: MobileNetV2 is selected as the spatial extractor due to the fact it is lightweight and quick. It divides normal convolutions into depthwise and pointwise components, reduced computation, and retaining good features.

Each video frame is processed independently by the network:

$$f_{spatial}(I_t) = \text{GlobalAvgPool}(\text{MobileNetV2}(I_t)) \in \mathbb{R}^{1280} \tag{7}$$

Frames in a set of video sequences $[B, T, C, H, W]$ are reformed to the shape $[B \cdot T, C, H, W]$ to allow the CNN to

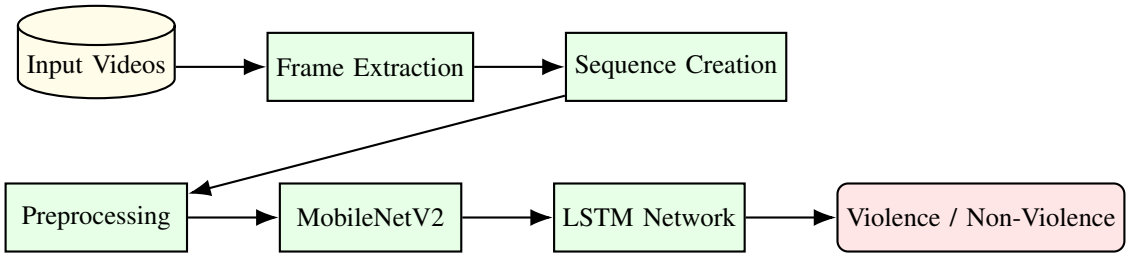

Fig. 2. Overview of the proposed CNN–LSTM based violence detection framework.

process them concurrently. This is followed by the restoration of the temporal structure.

$$\mathbf{F} = [f_{spatial}(I_0), \ldots, f_{spatial}(I_{15})] \in \mathbb{R}^{16\times 1280} \tag{8}$$

2) Transfer Learning Strategy: MobileNetV2 is initialised with ImageNet weight. During training, we can also freeze early layers, and adjust later layers to the patterns of violence. This increases the rate of learning and suppresses overfitting on the small data.

$$\theta = \{\theta_{frozen}, \theta_{trainable}\} \tag{9}$$

This transfer learning strategy improves convergence speed and reduces overfitting when training on the relatively limited surveillance dataset.

D. Temporal Modeling with LSTM

1) LSTM-Based Temporal Modeling: The extracted spatial feature sequence $\mathbf{F} \in \mathbb{R}^{T\times d}$, where $T = 16$ and $d = 1280$, is passed to a Long Short-Term Memory (LSTM) network to capture temporal dependencies across consecutive frames:

$$h_t = \text{LSTM}(f_t, h_{t-1}), \quad t = 1, \ldots, T \tag{10}$$

The LSTM allows the model to learn temporal patterns and variations, which are necessary to distinguish violent behavior and look-alike non-violent behavior.

2) LSTM Cell Dynamics: The LSTM cell maintains a memory that is determined by gates that determine what is to be retained or forgotten.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{11}$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{12}$$
$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{13}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{14}$$
$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{15}$$
$$h_t = o_t \odot \tanh(c_t) \tag{16}$$

E. Classification Head

The final hidden layer of the LSTM is passed to a dropout-based fully connected layer where it is then classified.

$$z_1 = \text{Dropout}(h_T, p = 0.5) \tag{17}$$
$$a_1 = \text{ReLU}(W_1 z_1 + b_1) \tag{18}$$
$$z_2 = \text{Dropout}(a_1, p = 0.5) \tag{19}$$
$$\text{logit} = W_2 z_2 + b_2 \tag{20}$$

The last logit is converted to a probability of the video being violent using a sigmoid.

$$P(\text{violence} \mid \mathbf{F}) = \sigma(\text{logit}) = \frac{1}{1 + e^{-\text{logit}}} \tag{21}$$

F. Training Strategy

1) Loss Function: The model is trained using binary cross-entropy loss, which is suitable for binary classification tasks such as violence and non-violence detection:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)] \tag{22}$$

where $y_i \in \{0, 1\}$ represents the ground truth label and $\hat{y}_i = \sigma(\text{logit}_i)$ denotes the predicted probability obtained after applying the sigmoid activation function.

2) Optimization: Model parameters are optimized using the Adam optimizer, which adapts the learning rate for each parameter during training:

$$\theta_{t+1} = \theta_t - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \tag{23}$$

Adam is used to tune the model weights. It varies the learning rate of individual weight in training.

$$\eta_t = \eta_0 \cdot \gamma^{\lfloor t/T_{decay} \rfloor} \tag{24}$$

We gradually decrease the learning rate with a schedule: initial rate is η_0 , factor of decay is γ , time of decay is T_{decay}.

G. Inference Strategy

1) Sequence-Based Prediction: We train 16-frame sequence videos during the inference process, just as the training. Every sequence runs across the trained CNN -LSTM to obtain a probability of violence.

$$\mathbf{S}_j = \{I_j, I_{j+1}, \ldots, I_{j+15}\} \tag{25}$$

This sequence-based inference approach ensures that temporal context is preserved during prediction and allows the model to analyze motion patterns across consecutive frames.

2) Decision Output: Every sequence receives a score in terms of violence chance. When the score is high, it is termed as violent and vice versa. This output may be used to assist in surveillance analysis or additional processing.

IV. Experimental Results

In this section, the experimental results of CNN-LSTM violence detection system are provided in the form of numbers and visual analysis of real tests.

A. Implementation Details

Our system was created in Python using PyTorch. Video reading and frame extraction was done by OpenCV. In all experiments, we tested and trained in a uniform controlled learning process under the same preprocessing, training, and evaluation procedure.

B. Dataset and Evaluation Protocol

The RWF-2000 dataset was done with experiments, this data set contains 2,000 real-world CCTV video clips, evenly split, with half violent and half non-violent. The standard dataset protocol was used to divide the dataset into training and testing subsets. Video frames were segmented in a constant sampling rate and divided into 16 frame sequences that were fed to the CNN -LSTM model both in training and inference.

C. Classification Performance

The proposed model was tested on the test set in terms of classification performance. The confusion matrix of the predictions is presented in Figure 3. The model is shown to perform well in both classes with the correct identification of a high number of both violent and non violent samples.

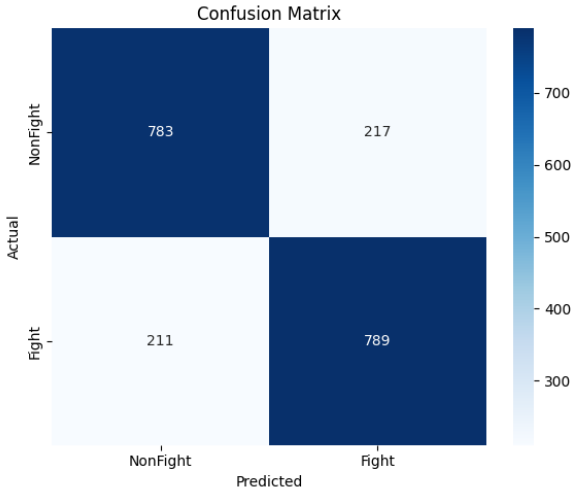

Fig. 3. Confusion matrix obtained on the test set.

The similarity of values in the diagonal of the confusion matrix shows that there is no high level of bias on the model on one class or another. False identifications are mostly made in situations that are hard to see with the eye like high-speed or partial blockage, which is prevalent with surveillance videos.

D. ROC Curve Analysis

The figure of receiver operating Characteristic (ROC) curve of the proposed approach is shown in figure 4. The AUC of about 0.87 reflects great capability in discrimination between violent video sequences and non-violent video sequences with different decision thresholds.

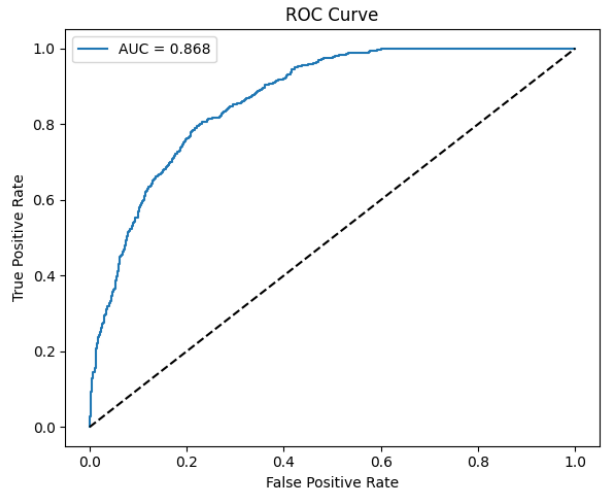

Fig. 4. ROC curve illustrating the trade-off between true positive rate and false positive rate.

The ROC analysis obliterates the fact that the learned spatiotemporal features provide reliable separation of the two classes even in the difficult real-life conditions.

E. Training and Validation Behavior

Figure 5 and 6 show the trend of training and validation losses and accuracy with respect to epochs. The accuracy of the training is increasing steadily with the acquired discriminative features in the model, and the relative accuracy of the validation does not change significantly during training.

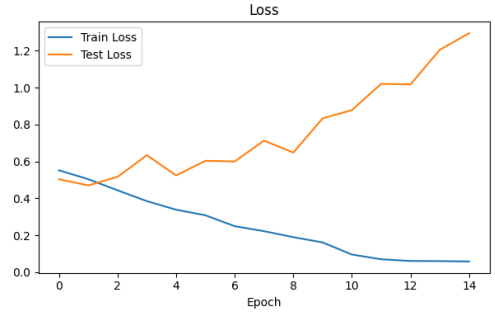

Fig. 5. Training and validation loss across epochs.

Despite the fact that a disconnect is found between training and validation accuracy, the validation performance is not

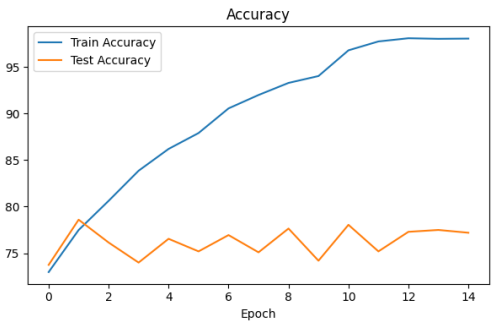

Fig. 6. Training and validation accuracy across epochs.

significantly reduced, implying that the model is able to extrapolate to missed data. This is typical of training deep learning models using actual surveillance data in the real world, which needs to be of high visual and motion variance.

F. Qualitative Results

Qualitative prediction results of the proposed model are depicted in figure 7. The system detects the violent action in surveillance, and the difference between the violent and non-violent crowds appropriately, which implies proper spatiotemporal knowledge.

Fig. 7. Qualitative results of the proposed system. Left: violent activity detected. Right: non-violent scenario correctly classified.

V. Discussion

A. Performance Analysis

The experimental findings show that the presented CNN-LSTM architecture is useful in terms of violence detection in CCTV videos. The model obtains a total classification score of about 82% meaning that the learning of spatial and temporal features can be combined to ensure that there is discrimination between violent and non-violent video sequences. This observation is also supported by the ROC curve analysis where the AUC is approximately 0.87 meaning that the two classes are strongly separable.

The confusion matrix analysis indicates that there is a balanced performance of the model between violent and non-violent classes, which implies that the model is not strongly biased to either category. Majority of misclassifications are made in a situation that is visually ambiguous, e.g., a scene involving rapid motion or partial blocking which are typical in the real world of surveillance.

B. Architectural Design Choices

MobileNetV2 as the spatial feature extractor is important in ensuring that the computational efficiency is maintained and at the same time the discriminative visual features are extracted. In comparison to the heavier convolutional network, MobileNetV2 offers an appropriate trade-off in accuracy and the complexity of the model, which is why it is applicable in the context of video-based surveillance analysis.

The LSTM network allows the temporal modeling of the image, which helps the system to capture motion patterns and temporal evolution based on successive frames. This temporal background plays an essential role in the separation of violent behavior and that which can be visually similar non-violent behavior which cannot be effectively discerned by viewing a single frame. Fixed length frame sequences are used to ensure that the model is able to learn motion continuity and retain a constant input structure.

C. Transfer Learning Impact

The pretrained weights of CNN using ImageNet are better to start with, in cases of limited surveillance data, better feature representation and stabilize the training process. Despite the fact that ImageNet pictures are very different to CCTV footage, the pretrained model has very strong low-level and mid-level features, i.e. edges and texture patterns, that can be generalized and be applied to violence detection tasks. Later refinement of finer layers allows adapting to domain specific patterns of motion and interaction.

D. Limitations

Although the proposed approach has promoted performance, it is limited in a number of ways. First, the model might not perform well in difficult environmental situations e.g. when there is a lack of light, a congested scene and when there is a considerable amount of occlusions. These conditions may cause visual illusions and bring down the reliability of detection.

Second, the fixed-length sequence approach can be insensitive to violent events which can be found at the frames sequence boundaries. Although overlapping sequences can be used to reduce this problem, temporal boundary effects can still affect consistency of detection.

Also, the present system conducts binary classification and fails to draw a line between various types and levels of severity of violence. These differences may be useful in prioritizing response in actual surveillance situations.

E. False Positive Observations

Once misclassified samples are inspected qualitatively it has been found that false positives are frequent due to non-violent actions that involve rapid or exaggerated movements. These are sports activities, dancing, unexpected movement of cameras, and interactions with a dense crowd. Such observations indicate that it may be possible to decrease the false alarms by adding more contextual cues or refinement strategies of motion.

F. Future Research Directions

Future studies can discuss the attention based mechanisms that pay attention to the salient regions and frames related to violent actions. Multi-task learning along with auxiliary tasks like person detection or pose estimation would be another way of improving robustness. Testing the model with different datasets would also give a better understanding of the overall generalization of the model. Lastly, visual explanations of models might enhance the trust and usability of real-life surveillance systems.

VI. Conclusion

In this paper, a deep learning algorithm to detect violence in CCTV footage is presented with the help of CNN and LSTM. It becomes familiar with the geographical locations and time, hence being able to differentiate between violent action and ordinary action on the real footage.

It makes use of MobileNetV2 to extract spatial features of each frame and uses LSTM to track variations over time. The analysis of the RWF-2000 data has shown an accuracy of approximately 82% and a ROC-AUC of 0.87 indicating that timing is of more importance than single-frame analysis.

Initializing ImageNet trained weights facilitates the model to train easier and acquire more features despite the fact that real CCTV video differs with normal pictures. Frame sequences allow the model to acquire patterns of movement of violent behaviors during a regular training program.

Although the technique is a good one, it is not without issues. It may not perform well with large crowds, dense occlusiveness or poor light. It also just mentions violent or not violent and cannot specify the type of violence and the severity which should be analyzed further.

Additional measures might include paying attention to key points, learning by multi-task and additional visual cues, and cross testing to observe generalization. Other options of the research could be to make the system more explainable through visual tools and give more detailed categories of violence.

Overall, the paper reveals that CNNLSTM is an effective model to identify violence in CCTV recording providing a promising foundation to the future evolution of smart surveillance.

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